Causal Inference with Machine Learning: Final Project

The main deliverable in this course is a final project.

Project
At a high level, the task for the final project is to find some causal question you're interested in, a dataset that may be able to answer it, and then use the tools from the course to conduct an analysis. A typical project might look something like:

1. Find one or more published papers estimating a causal effect of interest, where the estimate relies (implicitly or explicitly) on some detailed parametric assumption.

2. Carefully redo the analysis using general machine learning tools.

3. Write a report explaining your findings, and whether there was any change from more general analysis.

For such projects, you'll need to compute point estimates and confidence intervals for the effects of interest.
If there's some currently unpublished effect and data you're interested in: that's even better!
You'll be evaluated on the quality of your analysis and presentation. With respect to the analysis, you should make any causal identification assumptions explicit and carefully consider their plausibility; check any for quality of fit of the models you use; check any testable assumptions of the method (e.g., overlap in adjustment or instrument relevance in IV). Ideally, you should also conduct analysis of senstivity of your model to the choice of nuisance function model, and to possible violations of any causal identification assumptions.

Structure and Dates
Reports should be 4-8 pages, though you can have unlimited appendix space (e.g., for additional figures). Projects must be done in groups of 2 or 3 (subject to change if the number of students changes significantly at the drop deadline). There are two major dates: on **May 7** submit a 1 page writeup sketching the causal question of interest, the dataset you plan to analyze, and a plan for the analysis. The final projects are due **May 29** by midnight.

Data sources
You can get data from anywhere you please. Some convenient sources include:

1. Most top economics journals now require data to be published with articles (you can find a list here: `https://ideas.repec.org/top/top.journals.all.html`).

2. Harvard Data Repository: `https://dataverse.harvard.edu/`

3. `https://github.com/gsbDBI/ExperimentData`